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Studierichting: Informatica
Oefeningenreeks 3A nr. 12.2

$$f(x) = (x^2 + 5x - 6)(x^2 - 2x + 4)$$

product uitwerken:

$$f(x) = x^4 - 2x^3 + 4x^2 + 5x^3 - 10x^2 + 20x - 6x^2 + 12x - 24$$

$$= x^4 + 3x^3 - 12x^2 + 32x$$

$$f'(x) = 4x^3 + 9x^2 - 24x + 32$$

formule:

$$f'(x) = (2x + 5)(x^2 - 2x + 4) + (2x - 2)(x^2 + 5x - 6)$$

$$= 2x^3 - 4x^2 + 8x + 5x^2 - 10x + 20 + 2x^3 + 10x^2 - 12x - 2x^2 - 10x + 12$$

$$= 4x^3 + 9x^2 - 24x + 32$$

References